

A culture of the bacterium *Clostridium botulinum*, which produces the botulinum toxin. The toxin was approved for medical use in the USA in 1988.

IN JUNE 1988, JAMES HANSEN (b.1941) the director of NASA's Goddard Institute for Space Studies, in New York, reported to the US Senate Committee on Energy and Natural Resources that the average **global temperature was increasing** above what would be expected by normal climate variation. He noted that the world's temperature was greater than it had ever been since systematic recording had begun about a hundred years earlier. He said raised temperatures would probably cause an **increase in heat waves** and other **extreme weather events**. Importantly, he suggested that the main cause of the warming was the greenhouse effect (see pp.326–27), which was

being enhanced by the enormous amounts of **carbon dioxide released into the atmosphere** by the burning of fossil fuels.

Climate scientists were already well aware of global warming, and the possible challenges the world might face if the warming continued. In 1986, the World Meteorological Organization and the United Nations Environment Program had set up a body to examine the phenomenon: the **Advisory Group on Greenhouse Gases**. This small group was superseded by the UN's **Intergovernmental Panel on Climate Change (IPCC)**, which was formed in late 1988. The IPCC's first assessment report would be published two years later (see 1990).

“...**GLOBAL WARMING HAS REACHED A LEVEL SUCH THAT WE CAN ASCRIBE... A CAUSE AND EFFECT RELATIONSHIP BETWEEN THE GREENHOUSE EFFECT AND THE OBSERVED WARMING.**”

James Hansen, American climate scientist, testifying before the US Senate Committee on Energy and Natural Resources, 1988

In November, Dutch computer scientist Piet Beertema (b.1943) initiated a connection to the **US National Science Foundation Network (NSFnet)**. The NSFnet was a nationwide set of interconnected computer networks for academics. It formed the **backbone of the early Internet**. Other

organizations in the Netherlands and across Europe became connected soon afterward. Fortunately for computer scientists in the Netherlands, their connection to NSFnet was made two weeks after the **first internet worm** was released. The Morris worm—written by student Robert Morris at Cornell University, Ithaca, New York—**infected several thousand computers** across NSFnet, slowing them down. The cost in terms of downtime and the effort involved in removing the worm from infected computers is not known, but is estimated to be around \$1 million. Morris was convicted under the US Computer Fraud and Abuse Act.

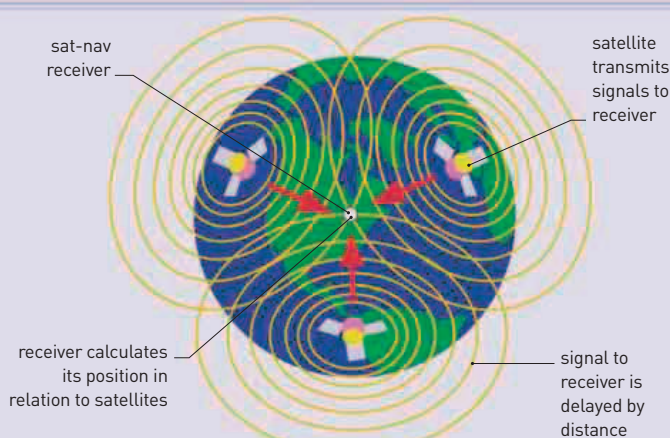
In February 1989, the first in a new phase of satellites was launched to **modernize the Global Positioning System (GPS)**, (see 1978). Over the next decade, 18 new satellites were placed in orbit. Since GPS was

initially a US military enterprise, access to highly accurate GPS signals was restricted to the armed services—mainly to ensure that military enemies would not be able to benefit. By the end of the 1990s, this restriction was lifted, and the general public was given access to the full service, **opening up a new market in navigational devices**—including in-car “satellite navigation” devices and GPS-enabled mobile phones. **Russia has a similar satellite navigation system**—the Globalnaya Navigatsionnaya Sputnikovaya Sistema (GLONASS), whose roots also lie in the 1970s. It became fully operational in 1993. Since the late 2000s, many satellite navigation devices use both GPS and GLONASS.

In December, botulinum toxin, **better known as “botox,”** was approved in the US for problems associated with eye muscles. The

HOW SATELLITE NAVIGATION WORKS

Satellite navigation (sat-nav) devices pick up signals sent out by satellites in orbit around Earth. Each satellite carries a very accurate atomic clock. From the difference between the time signals that are received from three or more satellites, a sat-nav device can calculate how far it is from each satellite. Using those distances, the device can work out its geographic position with great accuracy.



April 1, 1988 Publication of Stephen Hawking's hugely popular book *A Brief History of Time*

May 16, 1988 US Surgeon General declares nicotine as addictive as heroin or cocaine

June 23, 1988 James Hansen raises awareness of the term “global warming”

November 2, 1988 The Morris Worm becomes the first computer virus to infect computers across the Internet

November 1988 The Netherlands becomes the second country connected to the Internet

December 6, 1988 United Nations establishes the Intergovernmental Panel on Climate Change (IPCC)